

0.1 Inner Comparison

The cross-method and matched-fit results establish external effectiveness and isolate the contribution of teacher-conditioned deflection. We next hold the backbone, rendered inputs, pseudo labels, and evaluation protocol fixed, and visualize three narrower claims: whether the native numeric channel remains limiting under stronger prompting controls; whether the gain concentrates in teacher-human disagreement regimes; and whether the learned context-attention pattern is functionally responsible for the score. Exact numerical values are retained in the appendix tables.

0.1.1 Native Scoring Channel

All prompting controls use the same frozen backbone, multimodal input, and evaluation rubric. They differ only in how the native language channel is elicited. Figure 1 separates human-aligned performance from score resolution rather than compressing both into one multi-axis plot.

Figure 1 shows a clean separation between prompt engineering and learned readout. Analyze-then-score is the strongest native-channel control at 57.6% pairwise accuracy, but the plain frozen readout reaches 66.8%, and nuGuidance reaches 70.3%. The same separation appears in score resolution: nuGuidance exposes 341 effective levels, 14.4 $\times$ the 23.7 levels obtained by the strongest prompting control. Thus, better prompting improves neither human alignment nor score granularity enough to close the readout gap.

The stability result in Fig. 2 rules out the simpler explanation that the direct-prompt baseline merely used one brittle wording. Ensembling reduces sensitivity, but it does not recover the accuracy or score resolution of the learned readout. Together, the three plots support a channel bottleneck: the frozen representation can support substantially better scoring than the native textual number interface reports.

0.1.2 Semantic Disagreement Regimes

Overall MAE can conceal whether an evaluator succeeds on the cases that motivate semantic scoring. We therefore partition WOD-E2E using the fixed Scheme-B teacher score, public rule outputs, and existing human ratings, with no model-dependent manual case selection.

The regime breakdown in Fig. 3 distinguishes semantic correction from ordinary teacher fitting. On the agreement subset, nuGuidance is close to the teacher (1.61 versus 1.58 MAE), so its overall gain is not obtained by perturbing already-correct cases. The advantage appears where context is necessary: over-penalty MAE falls from 3.42 for the teacher to 2.42, and rule-pass blind-spot MAE falls from 3.95 to 2.79. The plain readout remains at 2.88 and 3.41, respectively, showing that the hard-case benefit is not explained by pseudo-score distillation alone.

Figure 4 provides the direct teacher-inheritance test. nuGuidance improves TranscendRate by 8.3 percentage points over the plain readout and by 4.6 points over LoRA.

This result supports, but does not assume, weak-to-strong behavior: the student more often moves toward human judgment on the teacher’s identifiable failure regime. If this pattern does not survive confidence intervals or the locked-test protocol, the conclusion must be narrowed to faithful frozen pseudo-score distillation.

0.1.3 Causal Readout Interventions

We intervene on the exact context-attention branch used by the filtering criterion. One trained nuGuidance checkpoint and scalar head are held fixed, and no intervention is followed by retraining. Figure 5 reports changes relative to the unchanged readout, which is the relevant paired causal comparison.

The intervention pattern is consistent across the external metric and the internal statistic. Uniform attention causes the largest accuracy decline (−9.4 points), top-attended suppression causes an −8.5-point decline, and within-modality shuffling causes an intermediate −5.2-point decline. Equal attention mass removed at random costs only −1.4 points. The decisive contrast is therefore 7.1 percentage points between top-attended and random suppression, accompanied by a 5.3-point difference in the deflection drop. Because the checkpoint and scalar head never change, this comparison supports a functional role for the learned context-attention pattern rather than a post-hoc visualization of an otherwise head-dominated scorer.

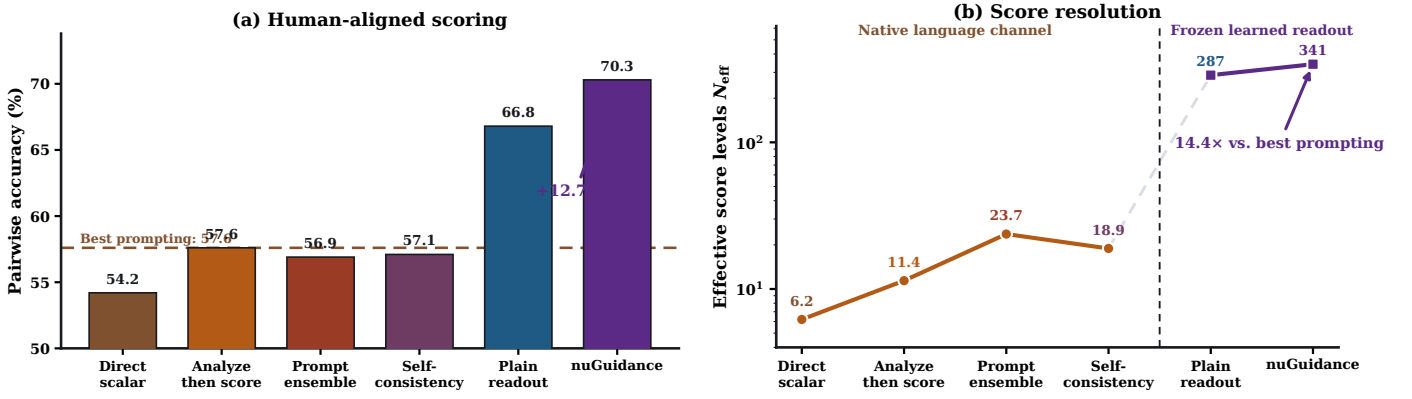


Figure 1: **Native textual scoring versus frozen learned readout.** Left: WOD-E2E pairwise accuracy. Right: effective score resolution N_{eff} on a logarithmic scale. Stronger prompting, paraphrase ensembling, and repeated sampling remain below the learned readouts. nuGuidance reaches 70.3% pairwise accuracy and 341 effective score levels, compared with 57.6% and 23.7 for the strongest prompting controls.

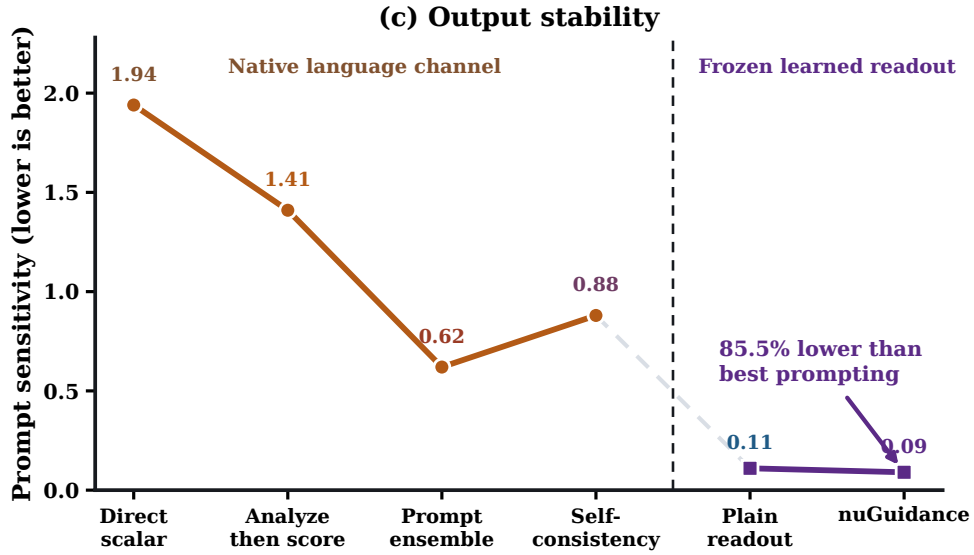


Figure 2: **Prompt sensitivity of the native channel.** The first four points are textual scoring controls; the final two are frozen learned readouts. Lower is better. The best prompting control remains at 0.62 sensitivity, whereas nuGuidance reaches 0.09, an 85.5% reduction.

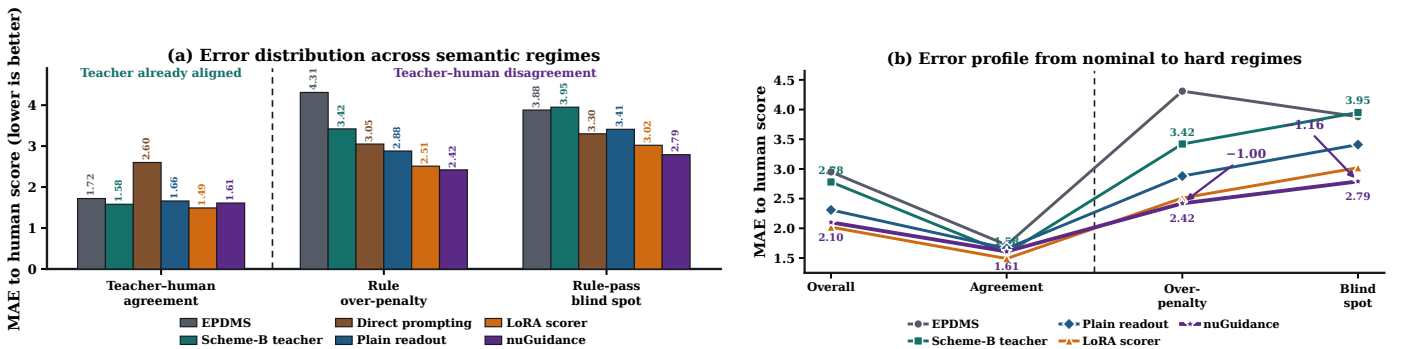


Figure 3: **Human-alignment error across semantic regimes.** Left: grouped MAE comparison on teacher-human agreement, rule over-penalty, and rule-pass blind-spot subsets. Right: the same methods traced from overall performance to the increasingly difficult semantic regimes. nuGuidance reduces teacher MAE by 1.00 point on rule over-penalty cases and by 1.16 points on rule-pass blind spots.

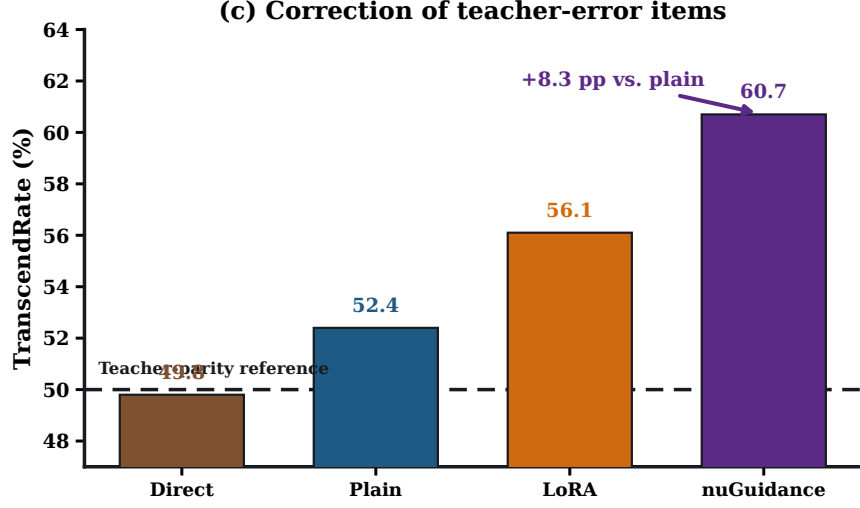


Figure 4: **Correction of teacher-error items.** TranscendRate is the proportion of disagreement items on which the student is closer to the existing human score than the Scheme-B teacher. The dashed 50% line is the teacher-parity reference. nuGuidance reaches 60.7%, compared with 52.4% for the plain readout and 56.1% for LoRA.

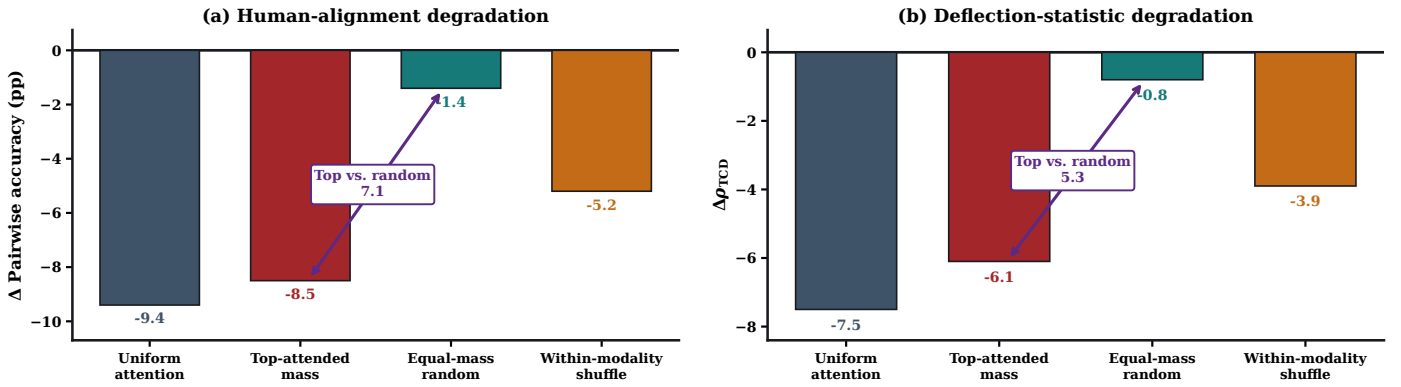


Figure 5: **Causal interventions on the learned attention readout.** Left: change in WOD-E2E pairwise accuracy. Right: change in held-out ρ_{TCD} . Suppressing top-attended context mass costs 8.5 accuracy points, compared with 1.4 points for equal-mass random suppression; the corresponding deflection drops are 6.1 and 0.8. Every bar is measured relative to the same unchanged checkpoint.